Akshaj Kashyap

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Education

University of California, Santa Barbara — Santa Barbara, CA
B.S. Computer Science | GPA: 3.76 — *Expected June 2028*
- **Coursework:** Operating Systems, Computer Networks, Advanced Application Programming, Computer Architecture, Systems Programming, Data Structures & Algorithms

Experience

PLAXCO Lab — Aug. 2025 – Present
Undergraduate Researcher — *Santa Barbara, CA*
- Analyzed **44 electrochemical exports across 11 datasets** in Python; generated concentration traces; investigating one- and two-compartment models.
- Continuing SACMES refactoring: reduced its main Python module from **5,268 to 4,834 lines**; passed **8 executed regression comparisons**, with additional cases skipped.

Sherwood Lab, UC Santa Barbara — Sept. 2026 – Present
Undergraduate Researcher | Active GPU Memory and Interconnect — *Santa Barbara, CA*
- Beginning GPU cache and memory-access research for AI workloads using **Hebrides, Barra, and Accel-Sim 2.0**, advised by Professor Tim Sherwood.

Stealth Startup — June 2026 – Sept. 2026
Technology Intern — *San Jose, CA*
- Built a **Python computer vision pipeline** integrating Roboflow, Segment Anything, and depth estimation to detect vegetation near cellular infrastructure for wildfire-risk screening.

Handshake AI Fellowship — Jan. 2026 – Present
AI Evaluation Fellow — *Remote*
- Evaluated **200+ AI-generated image edits** for realism and coherence, documenting structured written justifications.

Projects

CUDA Transformer Runtime | *C++20, CUDA, cuBLAS, CMake* | GitHub
- Built a **C++20/CUDA transformer inference runtime** with BPE tokenization, KV-cached decoding, and custom GPU kernels validated against independent CPU references.
- Passed **21/21 correctness tests** with maximum CPU/CUDA logit error of 2.56e-5; generated at **266.8 tokens/s** on a 15M-parameter model (RTX 3050 Laptop GPU, 64-token continuation).

C++ Matching Engine | *C++20, CMake, TCP, WAL* | GitHub
- Built a **price-time-priority order matcher** in C++20 with partial fills, write-ahead-log crash recovery, and bounded-queue TCP request handling.
- Used group commit to boost median passive-submit throughput **14x (250 to 3,498 ops/s)** versus a same-run one-record commit control (16 clients, local WSL2/ext4), preserving disk sync before responses.

MatchStream | *Python, Kafka/Redpanda, PostgreSQL, FastAPI* | GitHub
- Built a **durable event-processing pipeline** using Redpanda/Kafka, PostgreSQL, and FastAPI/WebSockets, enforcing per-match ordering, idempotent updates, and retry/DLQ handling.
- Replayed **3,549 real StatsBomb events** into durable match state, reproducing France's 4–3 win over Argentina; exercised broker/database outage recovery and WebSocket reconnection.

Leadership

SciTrek — Aug. 2025 – Present
Mentor — *Santa Barbara, CA*
- Helped develop a bioinformatics module used by 100+ K–12 students, translating technical concepts into clear lessons.

C.S.E.P. (F.U.S.E.) — Sep. 2025 – Present
Mentor — *Santa Barbara, CA*
- Selected as the only undergraduate mentor; helped develop and deliver a STEM module for 250+ students in 9+ classrooms.

Technical Skills

Languages: Python, C++, C, Java, SQL, TypeScript, JavaScript
Systems/Backend: CUDA, cuBLAS, CMake, TCP, PostgreSQL, Kafka/Redpanda, FastAPI, WebSockets
Tools/Platforms: Docker, GitHub Actions, Git, Linux